Final Project
Project Title
Network Monitoring and Performance Analysis System
Project Description

Students will design and develop an individual network monitoring application that can discover devices on a local network, monitor connectivity, analyze network performance, and generate diagnostic reports.

The project integrates major course topics including:

- OSI and TCP/IP Models
- IP Addressing and Routing
- TCP/UDP Protocols
- Network Performance Monitoring
- Network Security Concepts
- Python Network Programming
- Wireshark Traffic Analysis

The application should allow users to:

1. Scan and identify devices on a network.
2. Perform connectivity tests using ping.
3. Measure latency and packet loss.
4. Display network statistics.
5. Generate a summary report.
6. Provide basic security recommendations.

Students may develop the system using:

- Python
- Wireshark
- Socket Programming
- Cisco Packet Tracer (optional network simulation)
- Virtual Machines (optional)

This project directly supports the course objective of developing a practical networking application.

Project Objectives

Upon completion of the project, the student should be able to:

1. Apply networking concepts learned throughout the semester.
2. Use TCP/IP protocol knowledge in a practical application.
3. Analyze network performance metrics.
4. Demonstrate network troubleshooting skills.
5. Produce technical documentation.
6. Present and defend a networking solution.

Minimum Functional Requirements

Core Features (Required)

Feature 1: Network Discovery

- Detect local IP address
- Display network range
- List active devices

Feature 2: Connectivity Testing

- Ping selected devices
- Display response time
- Show packet loss percentage

Feature 3: Network Statistics

Display:

- Hostname
- IP Address
- Average Latency
- Maximum Latency
- Packet Loss

Feature 4: Network Report Generator

Generate a report containing:

- Date and time
- Devices detected
- Ping results
- Network performance summary

Feature 5: Security Recommendations

Examples:

- Weak connectivity alerts
- Unreachable hosts
- Basic firewall recommendations
- Password security reminders

Bonus Features (Optional)

Students may earn innovation points through:

- GUI using Tkinter
- Dashboard Charts
- Export to PDF
- Email Notification
- Port Scanner
- Wireshark Packet Integration
- Real-time Monitoring
- Cloud Database Storage

Required Deliverables

1. Project Proposal (Week 14)

Contents

- Title
- Problem Statement
- Objectives
- Methodology
- Tools and Technologies
- Expected Output

Length:

- 1 to 2 pages

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2. Progress Report (Week 15)

Include:

- System Design
- Screenshots
- Sample Code
- Accomplishments
- Remaining Tasks

Length:

- 2 to 3 pages

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3. Final Technical Report

Length:

- 8 to 12 pages

Format

Chapter 1: Introduction

- Background
- Problem Statement
- Objectives

Chapter 2: System Design

- Network Architecture
- Flowchart
- Use Case Diagram

Chapter 3: Implementation

- Tools Used
- Source Code Description

Chapter 4: Results and Testing

- Screenshots
- Performance Evaluation

Chapter 5: Conclusions and Recommendations

References

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4. Source Code
Must include:

- Commented code
- ReadMe File
- Installation Instructions

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5. Presentation Slides
10 to 15 Slides
Suggested Outline:

1. Title
2. Project Overview
3. Problem
4. Objectives
5. Methodology
6. System Architecture
7. Features
8. Demonstration
9. Results
10. Conclusion

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6. Demonstration Video
Duration:

- 5 to 10 minutes (should be in formal attire)

Must show:

- Application startup
- Feature demonstration
- Testing results
- Final output

Project Timeline

Week	Activity
Week 14	Proposal Submission
Week 15	Progress Report – Nov. 17
Week 16	Final Project Submission – Dec. 4
Week 16	Presentation and Demonstration – Dec. 8-10

Individual Project Grading Matrix

Criteria	Description	Points
Functionality	System works correctly and meets requirements	30
Technical Implementation	Proper use of networking concepts, protocols, and tools	20

User Interface and Design	Professional appearance and usability	10
Documentation	Quality of report, diagrams, and technical writing	15
Testing and Evaluation	Evidence of testing and analysis	10
Presentation and Demonstration	Clear explanation and project defense	10
Innovation and Creativity	Additional unique features	5
TOTAL		**100**

Detailed Performance Levels

Rating	Percentage
Excellent	90–100
Very Good	80–89
Good	70–79
Fair	60–69
Poor	Below 60

Functionality Rubric (30 Points)

Level	Description	Points
Excellent	All requirements implemented and working	27–30
Very Good	Minor issues present	23–26
Good	Most requirements working	19–22
Fair	Several incomplete functions	15–18
Poor	Major functionality missing	0–14

Technical Implementation Rubric (20 Points)

Level	Description	Points
Excellent	Demonstrates excellent networking knowledge	18–20
Very Good	Appropriate use of protocols and tools	15–17
Good	Adequate implementation	12–14
Fair	Limited networking integration	8–11
Poor	Weak application of concepts	0–7

Submission Requirements
Students must submit a compressed folder named:
LASTNAME_FIRSTNAME_FINALPROJECT.zip
Containing:
/SourceCode
/Documentation – exclude from zip file, to be submitted online via TEAMS
/Presentation
/VideoDemo
README.txt
Failure to submit any required component will result in deductions according to the grading rubric.

Expected Learning Outcomes
By completing the project, students will demonstrate the ability to:

- Design networking solutions.
- Implement network applications using Python.
- Analyze communication protocols.
- Troubleshoot network issues.
- Evaluate network performance.
- Document and present technical projects effectively.

This project aligns directly with the course objectives and Weeks 1–13 topics of the syllabus while remaining achievable as an **individual semester-long capstone project**.